

# Employee Attrition Prediction Using Machine Learning

## 1. Business Problem

Employee attrition can lead to increased recruitment costs, productivity loss, and disruption to business operations. This project aims to identify the key factors associated with employee attrition and develop a predictive model that can help organisations identify employees at higher risk of leaving. The findings may support HR teams in designing targeted retention strategies and improving workforce planning.

## 2. Dataset

- 1,470 employee records with 26 original variables. The dataset includes demographic, job-related, compensation, tenure, satisfaction, and performance information. **Attrition** was used as the binary target variable. Constant, identifier, and non-predictive variables were removed before modelling.

## 3. Project Workflow

The project followed a complete machine learning pipeline:

- **Data Understanding**
  - Examined dataset structure and feature types.
  - Assessed data quality, including missing values, duplicates, and class distribution.
- **Data Preparation**
  - Removed irrelevant and non-informative features.
  - Encoded categorical variables.
  - Split the data into training and test sets.
  - Standardised features for Logistic Regression.
- **Explore Data Analysis**
  - Analysed numeric and categorical features.
  - Applied statistical tests to validate relationships with attrition.
- **Model Development and Evaluation**
  - Developed Logistic Regression, Random Forest, and Gradient Boosting models.
  - Compared model performance and assessed generalisation.
  - Optimised the final Logistic Regression model using threshold tuning.

## 4. Key findings

- Employees working overtime had a notably higher attrition rate than those who did not.
- Employees who left tended to have lower monthly income, fewer total working years, shorter tenure with their current manager, and lower environment satisfaction.
- Frequent business travel and certain job roles were also associated with higher attrition.
- Overall, attrition was linked to a combination of workload, career stage, compensation, and workplace experience, rather than one single factor.

## 5. Model Comparison

Three machine learning models were developed and compared.

| Model               | Key findings                                         |
|---------------------|------------------------------------------------------|
| Logistic Regression | Best balance between recall, F1-score, and ROC-AUC   |
| Random forest       | High training performance but severe overfitting     |
| Gradient Boosting   | High accuracy but lower recall for employee who left |

Although Gradient Boosting achieved the highest accuracy. Logistic Regression demonstrated better overall discrimination and generation while identifying a larger proportion of employees at risk of attrition.

## 6. Final Model

Scaled Logistic Regression with threshold optimisation (0.6), selected for its balanced recall, F1-score, ROC-AUC, and strong generalisation performance.

## 7. Business Recommendations

- Monitor employees with frequent overtime and high travel requirements.
- Improve workplace satisfaction through employee engagement initiatives.
- Support career development and retention programmes for employees with shorter tenure.
- Use predictive analytics to identify employees who may benefit from proactive retention strategies.

## 8. Technologies & Methods

- **Programming:** Python
- **Data Analysis:** pandas, NumPy, matplotlib
- **Machine Learning**
  - scikit-learn
  - Logistic Regression

- Random Forest
- Gradient Boosting
- **Statistical Analysis**
  - Mann–Whitney U Test
  - Chi-Square Test
  - Point-Biserial Correlation